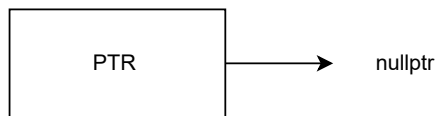
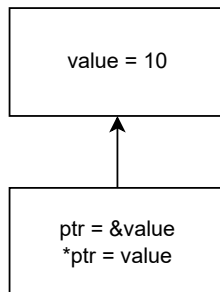


Seatwork 2.1: Memory Diagram Exercise

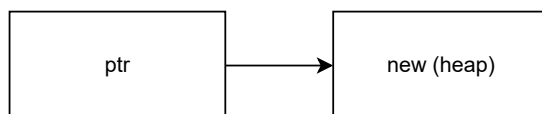
1. `int *ptr = nullptr;`



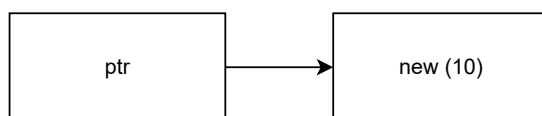
2. `int value = 10; int *ptr = &value;`



3. `int *ptr = new int;`



4. `int *ptr = new int(10);`



5. Differentiate Constructor, Destructor, Copy Constructor, Copy Assignment Operator.

The copy constructor is used when the source of the assignment is passed to the assignment operator.

A constructor is a special method that is automatically called when an object of a class is created.

A destructor is a special member function, prefixed with '~', that is automatically called when an object goes out of scope or is destroyed to free resources like memory, files, or connections.

A copy assignment operator is used with the "operator=" to create a new object from an existing one.